

Muzammil Nawaz Khan

Software Engineer

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SUMMARY

Software engineer with 3+ years of full-stack experience (frontend UI, Django backend, end-to-end features for an e-commerce business) and 1+ year building production AI services in Python. Built FastAPI services, data pipelines and an internal EDI viewer at CareCloud, and shipped full-stack projects with Django/DRF, React, Next.js, Docker and CI/CD. Comfortable taking features from design to deployment. Open to remote work and relocation.

SKILLS

Backend & Infrastructure: FastAPI, Django / DRF, REST APIs, Docker, Kubernetes, Git/GitHub, CI/CD

Frontend & Web: Next.js, React (Vite), REST APIs, WebSockets, SSE

Languages: Python, SQL, Rust, C++, JavaScript/TypeScript

Generative AI & Agents: OpenAI API, Anthropic API, LLMs, prompt engineering, structured outputs, RAG, LLM-as-judge evaluation, LangChain, LangGraph, CrewAI, AutoGen, Model Context Protocol (MCP), n8n, ChromaDB, FAISS, custom HNSW, BM25, Reciprocal Rank Fusion

Machine Learning: Python, PyTorch, TensorFlow, Keras, scikit-learn, CatBoost, XGBoost, AutoGluon, Hugging Face Transformers, Pandas, NumPy

Domain: Healthcare IT, medical imaging, AI training data

EXPERIENCE

Junior AI Engineer, CareCloud (MTBC)

Jul 2025 – Aug 2026

Islamabad, Pakistan

- Engineered Python pipelines that pull Hikvision NVR logs, clean them and load them to a database for HR attendance analytics; built an internal FastAPI EDI Viewer for inspecting raw healthcare claim data.
- Designed an n8n workflow acting as a Model Context Protocol (MCP) server for conversation state, backend prompts and call routing, wrapping APIs as tool calls. Cut inference costs 35% through prompt caching.
- Worked across a real-time voice AI front-desk agent on LiveKit (WebRTC) with ElevenLabs TTS, OpenAI models and Silero VAD for low-latency interruption handling; Twilio/SIP trunking for inbound calls with transfer to human staff; LangGraph for multi-state orchestration. Reduced front-desk handling time by 40%.
- Delivered a production claims-classification system (CatBoost, XGBoost, AutoGluon) at 95% parent-model accuracy and 92%+ category-level accuracy across 42,900+ cases, with daily feedback-loop retraining and multi-month tracking of accuracy and stability.

Software / AI Engineer, Suidhaga

Mar 2022 – Nov 2025

Remote, part-time alongside studies

- Progressed through the full engineering stack over 3+ years for an e-commerce business: built frontend UI, moved to Django-focused backend development, then owned full-stack web features end to end.
- Applied data analysis and data science to business data; later built ARIMA-family forecasting models to predict product prices.

Freelance AI Engineer, Independent

Sep 2022 – Present

Remote

- Delivered AI development projects for clients, including model training and applied AI application builds.

AI/ML Research Intern, Risetech

Jul 2024 – Aug 2024

Rawalpindi / Islamabad, Pakistan

- Benchmarked 3D U-Net against SegFormer3D and selected SegFormer3D (85%+ Dice). Used GradCAM for explainability and FastAPI for serving.

PROJECTS

ELD Trip Planner (Django, DRF, React/Vite, Leaflet, Nominatim, OSRM): Plans truck trips against FMCSA hours-of-service rules and draws daily ELD log sheets. [GitHub: github.com/muzammil5539/eld-trip-planner](https://github.com/muzammil5539/eld-trip-planner) | [Live: eld-trip-planner-sepia.vercel.app](https://eld-trip-planner-sepia.vercel.app)

Fuel Route Optimizer (Django, OSRM): Geocoded about 8,150 fuel stations, OSRM routing, an 8-mile corridor filter and a Dijkstra cheapest-fuel plan with a 500-mile range; regression-tested. [GitHub: github.com/muzammil5539/django_application](https://github.com/muzammil5539/django_application)

Conversational AI Agent (ReAct) (LangGraph, FastAPI, Next.js, OpenAI): Six tools (Calculator, Weather, DateTime, Web Search, Code Interpreter, RAG Search), visible reasoning traces, WebSocket streaming and a SQLite checkpoint for memory.

Document Summarizer (FastAPI, Next.js 14, LangChain, ChromaDB): Automatic strategy router (Stuff, Map-Reduce, Refine, Recursive), SSE token streaming and async pipelines. About 50% lower latency and 99% factual consistency.

Qadri Traders (E-commerce website): Online storefront built and deployed on Vercel. [Live: qadri-traders.vercel.app](https://qadri-traders.vercel.app)

EDUCATION

Bachelor of Computer Engineering, National University of Sciences and Technology (NUST) Nov 2021 – Jul 2025
Islamabad, Pakistan
CGPA 3.25. Final year project: Neurofusion, 3D brain tumor segmentation (SegFormer3D, BraTS20, 85%+ Dice); 1st Place COMPPEC 2025, Departmental Silver Medal.

F.Sc. Pre-Engineering, Ghazali Premier College Jun 2018 – Aug 2020
Lahore, Pakistan

CERTIFICATIONS

- IBM AI Engineering Professional Certificate
- IBM Generative AI Engineering Professional Certificate
- Large Language Model Operations (LLMOps) Specialization, Duke University
- Complete Modern C++ (C++11/14/17), Packt

HONORS & AWARDS

1st Place, COMPPEC 2025 | Departmental Silver Medal